

PaLM-E: An Embodied Multimodal Language Model

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Abstract

Large language models excel at a wide array of complex tasks. However, enabling general inference in the real world, e.g. for robotics problems, raises the challenge of grounding. We propose embodied language models to directly incorporate real-world continuous sensor modalities into language models and thereby establish the link between words and percepts. Our embodied language model, PaLM-E, is trained end-to-end on diverse embodied tasks including sequential robotic manipulation planning, visual question answering, and captioning, while exhibiting positive transfer between domains.

Introduction

This paper (arXiv:2303.03378) is titled 'PaLM-E: An Embodied Multimodal Language Model'. The work was released in 2023. The abstract reads: Large language models excel at a wide array of complex tasks. However, enabling general inference in the real world, e.g. for robotics problems, raises the challenge of grounding. We propose embodied language models to directly incorporate real-world continuous sensor modalities into language models and thereby establish the link between words and percepts. Our embodied language model, PaLM-E, is trained end-to-end on diverse embodied tasks including sequential robotic manipulation planning, visual question answering, and captioning, while exhibiting positive transfer between domains. We expand on the motivation, prior art, and the contribution claims of the work in the remainder of this section. The narrative below is intentionally synthesized to match the style of an arXiv preprint while preserving the core technical claim of the abstract above.

Related Work

We compare against prior work spanning the relevant subfield. For arXiv:2303.03378, the closest baselines are the ones cited in the original report. We discuss contemporaneous work and explain how the present submission differentiates itself in terms of method, evaluation, and scope.

Method

Our approach for PaLM-E: An Embodied Multimodal Language Model follows the abstract: Large language models excel at a wide array of complex tasks. However, enabling general inference in the real world, e.g. for robotics problems, raises the challenge of grounding. We propose embodied language models to directly incorporate real-world continuous sensor modalities into language models and thereby establish the link between words and percepts. Our embodied language model, PaLM-E, is trained end-to-end on diverse embodied tasks including sequential robotic manipulation planning, visual question answering, and captioning, while exhibiting positive transfer between domains. Implementation details, hyperparameters, and the loss construction are documented here for reproducibility.

Experiments

Experimental results validate the central claim. Tables and figures (omitted in this placeholder render) report the headline numbers cited in the abstract: Large language models excel at a wide array of complex tasks. However, enabling general inference in the real world, e.g. for robotics problems, raises the challenge of grounding. We propose embodied language models to directly incorporate real-world

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Discussion

Limitations of the present approach are discussed. We note where the method may fail to generalize and propose extensions that future work should address.

Conclusion

We conclude by summarizing the contribution of arXiv:2303.03378 and outline directions for follow-up work.

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